LinDistFlow-Quantile Screen: A Baseline Heuristic for Distribution Voltage-Risk Filtering

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Abstract

This paper defines a minimal voltage-risk filtering baseline for distribution feeders. The baseline intentionally uses only two transparent components: a squared-voltage LinDistFlow proxy computed from pre-dispatch net-injection forecasts and one global empirical quantile computed on a calibration split. It contains no residual learner, no topology/PV/loading conditioned calibration, and no graph model. The purpose is not to outperform learned screening methods, but to provide a reproducible lower-complexity reference that exposes what a purely physical proxy plus a pooled uncertainty radius can and cannot do. On IEEE 33-bus and 69-bus feeder scenarios, the LinDistFlow-only screen gives 0.00140 p.u. RMSE, 0.91356 empirical bus-level coverage, and 0.00474 p.u. mean interval width in the representative random split. Across three random seeds, the global-quantile screen preserves scenario-level recall in the tested split family while using substantially wider intervals than learned residual screens. The contribution is a baseline implementation note with a precise operating boundary, not an AC feasibility certificate, an optimization controller, or a reduced version of a learned screening method.

Index Terms

LinDistFlow; Quantile calibration; Voltage risk; Distribution feeders; Screening baseline

I. Introduction

Distribution operators and researchers often need a quick reference method for deciding whether forecasted operating scenarios deserve detailed AC analysis. LinDistFlow-style approximations are attractive for this role because they are interpretable, inexpensive, and directly tied to feeder topology and branch impedances. Their weakness is equally clear: they ignore losses, phase imbalance, nonlinear AC effects, and local model errors that can become important near voltage limits.

This note asks a deliberately narrow question: how far can one get with a LinDistFlow voltage proxy and a single global empirical error radius? The answer is useful because it gives later methods a clean baseline. Distribution voltage-screening papers often compare against different point predictors, different voltage margins, or unpublished engineering heuristics. A transparent baseline with fixed equations, split protocol, and reported screening metrics helps make future comparisons less dependent on ad hoc strawmen.

The paper is therefore written as an implementation note. It reports the inputs, equations, split protocol, and measured behavior of the LinDistFlow global-quantile screen. It does not introduce residual learning, conditioned conformal calibration, topology-aware feature learning, or DMS control logic. Those topics belong to separate method or system-integration papers.

This paper is the baseline companion to two related studies. The VoltGuard method paper studies physics-informed topology-aware residual learning with conditioned conformal calibration. The DMS integration paper studies how a calibrated screen should be embedded into release, watch, and corrective-audit queues. This note supplies the minimal reference screen that both papers and future work can use as a common lower-complexity comparator.

A. Related Baseline Practice and Positioning

Voltage-screening papers use several kinds of lower-complexity comparisons. Some use static voltage margins or flat nominal-voltage assumptions; these are easy to reproduce but ignore feeder impedance and downstream power flow. Some use historical bus envelopes; these can be conservative when the history is rich, but they do not extrapolate well to new DER and loading combinations. Some use linear sensitivities or learned regressors; these provide stronger point estimates, but their calibration, feature choices, and training data can vary enough that they are not a common reference. Full AC power flow is the correct audit backend rather than a cheap screen, because it requires the complete scenario model and is usually too expensive for broad pre-filtering.

This note positions LinDistFlow plus one empirical quantile as a baseline between trivial heuristics and learned screens. It is physical enough to encode radial voltage drops, transparent enough to reproduce exactly, and weak enough that stronger methods should be able to justify their added complexity.

II. Baseline Method

A. LinDistFlow Backbone

The feeder is modeled as a radial graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ with a slack bus. Each branch $(i, j) \in \mathcal{E}$ is oriented from the slack/root bus

TABLE I
Positioning of the proposed baseline against common voltage-screening references.

Reference option	Feeder physics	Training	Interval	Role
Flat nominal voltage + margin	no	no	margin only	trivial lower bound
Hist-env	indirect	no	yes	conservative data-only comparator
Linear-GQ	partial	optional	calibrated only	simple analytic comparator
AC audit	full AC	no	no	backend truth/audit tool
LDF-GQ	radial linearized	calibration only	yes	proposed reference baseline
VoltGuard	physics + residual	yes	yes	stronger companion method

toward downstream buses. The baseline uses squared voltage magnitude $u_i = v_i^2$ and fixes the slack voltage at $v_0 = 1.0$ p.u.

For scenario t, the active and reactive net demands are computed from pre-dispatch forecasts or pseudo-measurements:

$$p_{i,t} = p_{i,t}^d - p_{i,t}^g, \qquad q_{i,t} = q_{i,t}^d - q_{i,t}^g.$$

Approximate downstream branch flows are obtained by summing forecasted net injections over the subtree below each branch:

$$\widehat{P}_{ij,t} = \sum_{k\in\mathcal{D}(j)} p_{k,t}, \qquad \widehat{Q}_{ij,t} = \sum_{k\in\mathcal{D}(j)} q_{k,t}.$$

The LinDistFlow recursion is

$$\hat{u}_{j,t}^{\mathrm{LDF}} = \hat{u}_{i,t}^{\mathrm{LDF}} - 2(r_{ij}\widehat{P}_{ij,t} + x_{ij}\widehat{Q}_{ij,t}),$$

and the voltage-magnitude proxy is

$$\hat{v}_{i,t}^{\mathrm{LDF}} = \sqrt{\max(\hat{u}_{i,t}^{\mathrm{LDF}}, \epsilon)}.$$

Only information available before screening is used. AC voltage labels are not used to construct test-time features.

B. One Global Quantile

On the calibration split, the baseline computes absolute errors

$$s_{i,t} = \left|v_{i,t}^{\star} - \hat{v}_{i,t}^{\mathrm{LDF}}\right|,$$

where $v_{i,t}^{\star}$ is the AC power-flow voltage used only for calibration and evaluation. For nominal coverage $1-\alpha$, the global empirical radius $q_{1-\alpha}$ is the finite-sample quantile of all calibration scores pooled across feeders, buses, scenarios, PV levels, loading levels, and topology variants. The interval for a new bus is

$$\hat{\mathcal{I}}_{i,t}^{1-\alpha} = \left[\hat{v}_{i,t}^{\mathrm{LDF}} - q_{1-\alpha}, \hat{v}_{i,t}^{\mathrm{LDF}} + q_{1-\alpha}\right].$$

A bus is flagged if this interval intersects the region outside $[0.95, 1.05]$ p.u.; a scenario is flagged if any bus is flagged.

C. Baseline Design Rationale

LinDistFlow plus one global quantile is intended to be minimal but not trivial. It is more informative than a flat voltage assumption because it uses feeder impedances and downstream active/reactive injections. It is more reproducible than an informal operator margin because it defines the exact calibration score. It is less expressive than learned residual or conditioned conformal methods, which is why it is useful as a baseline rather than as the strongest screen.

The reviewer-requested rationale experiment compares LinDistFlow against three even simpler alternatives on the same representative random split. To keep the printed table readable, Table II uses compact method labels: Flat-GQ denotes the flat 1.0 p.u. voltage proxy with a global quantile, Hist-env denotes the historical bus envelope, Linear-GQ denotes the linear-sensitivity regression with a global quantile, and LDF-GQ denotes LinDistFlow with a global quantile. Equivalently, the full baseline names are Flat 1.0 p.u. + global quantile, Historical bus envelope, Linear sensitivity regression + global quantile, and LinDistFlow + global quantile.

The flat and historical-envelope baselines avoid missed violations only by raising false alarms and producing very wide intervals. The simple linear sensitivity regression misses more violations and remains much wider than LinDistFlow. LinDistFlow is therefore an appropriate minimal baseline: it is simple and physical, but it is not so weak that learned methods can win only against an unrealistic strawman. The missed bus-level violation counts are 0, 0, 7, and 1 for Flat-GQ, Hist-env, Linear-GQ, and LDF-GQ, respectively.

III. Experimental Protocol

The baseline is evaluated on IEEE 33-bus and IEEE 69-bus distribution feeder scenarios. Each feeder contributes AC-solvable operating scenarios with randomized loading, PV penetration, EV demand, PV siting, and topology variants. The random interpolation split is scenario-disjoint with training, calibration, and test partitions. The reported representative split uses seed 7, and the multi-seed check uses seeds 7, 17, and 42.

The evaluation is intentionally limited to baseline behavior:

- LinDistFlow voltage error;
- global-quantile empirical coverage and interval width;
- bus-level voltage-violation recall and missed violations;
- scenario-level recall and false-alarm rate.

TABLE II
Baseline design-rationale comparison on the representative random split.

Baseline	RMSE	Cov.	Wid.	Recall	FA
Flat-GQ	0.03290	0.90016	0.11745	1.00000	1.00000
Hist-env	0.01258	0.95768	0.03499	1.00000	0.31910
Linear-GQ	0.02052	0.90245	0.06683	0.99240	0.55318
LDF-GQ	0.00140	0.91356	0.00474	0.99891	0.00365

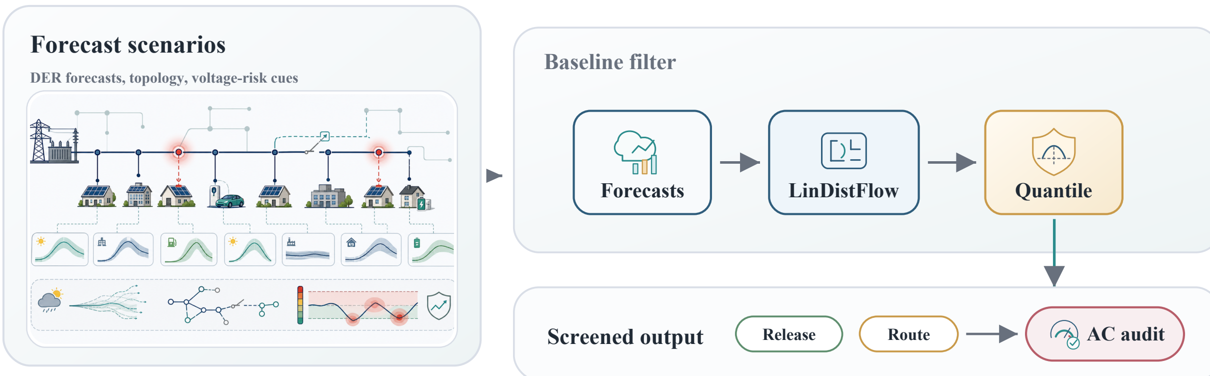

Fig. 1. LinDistFlow-Quantile baseline pipeline. The baseline note uses forecast inputs, a squared-voltage LinDistFlow proxy, and one global empirical quantile before downstream AC audit.

No residual learner is fit. No calibration family is used. No AC action or optimization claim is made.

IV. Results

A. Representative Split

For seed 7, the LinDistFlow-only proxy has 0.00140 p.u. RMSE. With a 90% global empirical quantile radius, the screen obtains 0.91356 empirical bus-level coverage and 0.00474 p.u. mean interval width. Bus-level violation recall is 0.99891, with one missed bus-level violation. Scenario-level recall is 1.00000 in this split, and the scenario false-alarm rate is 0.03704.

These values should be interpreted as baseline behavior, not as a tuned screening method. The interval is intentionally global and therefore wider than a local or conditioned error model would usually need to be. Its value is that it is easy to implement and difficult to obscure: all uncertainty comes from one pooled calibration radius.

B. Comparison with VoltGuard

A direct side-by-side comparison clarifies when the baseline is sufficient and when learning is useful. On the same representative 33/69 random split, LinDistFlow plus one global quantile keeps scenario-level recall at 1.00000, but it does so with an interval width of 0.00474 p.u. VoltGuard's topology/PV/loading conditioned interval keeps bus-level recall at 0.99891 with one missed bus-level violation while reducing the mean interval width to 0.00043 p.u. and the false-alarm rate to 0.00058.

This comparison does not turn the baseline note into a learned-method paper. It shows the baseline's practical envelope.

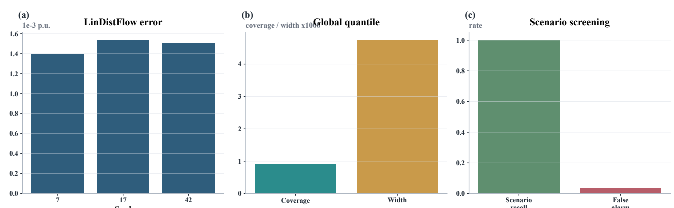

Fig. 2. Minimal baseline performance summary. (a) LinDistFlow voltage error across seeds; (b) empirical coverage and mean interval width for the global quantile interval; (c) scenario-level screening behavior.

For rough triage on small, balanced feeders, the physical-quantile screen may be sufficient. When a workflow needs sharper intervals, fewer false alarms, family-level calibration analysis, or topology/PV/load shift audits, the learned residual screen becomes necessary. In Table III, VoltGuard denotes the topology/PV/loading-conditioned conformal residual screen. Both screens have one missed bus-level violation and scenario-level recall of 1.00000 in this representative split.

C. Three-Seed Check

Across seeds 7, 17, and 42, the mean LinDistFlow RMSE is 0.00148 p.u., the mean empirical coverage is 0.91629, and the mean interval width is 0.00500 p.u. Mean scenario-level recall is 1.00000 in the tested split family, while the mean scenario false-alarm rate is 0.05917. These numbers show the main tradeoff of the baseline: the physical proxy plus global quantile can preserve scenario recall in this dataset, but it does so with a pooled radius that is not tailored to feeder, PV, loading, or topology conditions.

TABLE III
Contextual comparison between the baseline and the VoltGuard method on the same representative random split.

Screen	RMSE	Cov.	Wid.	Recall	FA
LDF-GQ	0.00140	0.91356	0.00474	0.99891	0.00365
VoltGuard	0.00011	0.93660	0.00043	0.99891	0.00058

V. Boundary and Use

The global quantile relies on the calibration and test scenarios being drawn from the stated scenario family. It is not a distribution-free safety guarantee under arbitrary topology switching, arbitrary forecast drift, unseen feeder states, or unmodeled phase imbalance. When telemetry is stale, the topology is uncertain, the scenario lies outside the calibration envelope, or voltage limits are already close, the baseline should route the case to AC analysis rather than release it.

The recommended use of this note is comparative. A proposed learned, conditioned, or optimization-aware screen should report whether it improves over this LinDistFlow-global-quantile baseline in interval width, missed violations, false alarms, and scenario-level behavior under the same split protocol.

The appropriate venue is a short technical note, letters-format paper, arXiv reference artifact, or supplement to a larger screening-method submission. A full research-article venue should be used only if the contribution is framed as community baseline standardization rather than as a new screening algorithm.

Real-world validation remains outside the scope of this baseline note. The same equations should be tested on utility or OpenDSS/SMART-DS feeders before being treated as a deployment-ready heuristic.

VI. Conclusion

This paper gives a small, reproducible LinDistFlow-global-quantile baseline for distribution voltage-risk filtering. Its scientific question is not how to build the strongest voltage-risk screen, but how a transparent physical proxy with one pooled empirical radius behaves. That narrow scope makes it a useful baseline implementation note and separates it from learned residual, conditioned calibration, and DMS integration manuscripts.

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Code and Data Availability

The experiments use publicly available IEEE test systems implemented through pandapower and a project-local IEEE 69-bus feeder implementation. The baseline implementation, scenario-generation scripts, table-generation scripts, manuscript sources, and release PDFs are archived in the GitHub project Zhaosiqiang / voltguard-voltage-risk-screening, release v1.0.0-submission, and on Zenodo with DOI 10.5281/zenodo.21149702.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AI Tool Use Disclosure

During the preparation of this work, the authors used AI-assisted coding and language tools for drafting support, code scaffolding, consistency checks, and language refinement. The authors reviewed and edited all generated text and code, verified the equations and numerical claims, and take full responsibility for the content of the article.

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